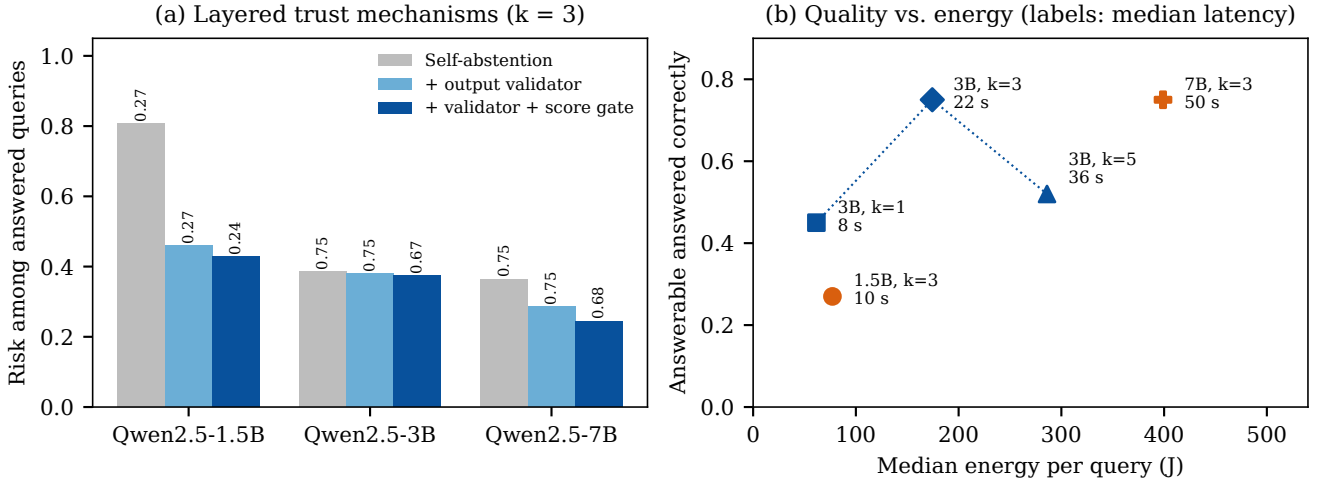
(a) Layered trust mechanisms (k = 3)
Risk among answered queries
Self-abstention
+ output validator
+ validator + score gate
0.27
0.27
0.24
0.75
0.75
0.67
0.75
0.75
0.68
Qwen2.5-1.5B
Qwen2.5-3B
Qwen2.5-7B
(b) Quality vs. energy (labels: median latency)
Answerable answered correctly
3B, k=3
22 s
7B, k=3
50 s
3B, k=5
36 s
3B, k=1
8 s
1.5B, k=3
10 s
Median energy per query (J)